

CME 222

MASS TRANSFER

Diffusive Mass Transfer

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Estimation of Diffusion Coefficient (Liquid Diffusivity)

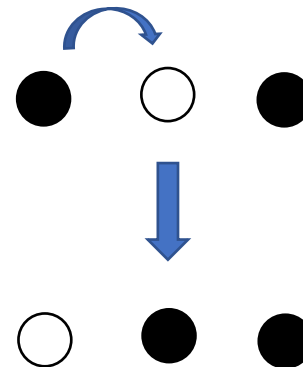
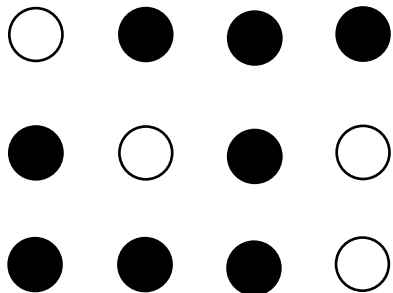
Liquid Mass Diffusivity

- Theory of diffusion is not as advanced, less experimental data
- Diffusivities in liquids are 4-5 order of magnitude smaller than gases
- Can diffuse as molecules or ions (Ex: NaCl)

Liquid Mass Diffusivity

Eyring “hole” Theory

- The liquid is treated as quasi-crystalline lattice model interspersed with holes
- The transport involves the jumping of solute molecules into the holes within the lattice model



Liquid Mass Diffusivity

Hydrodynamical Theory

- Liquid diffusion coefficient is related to solute molecules' mobility
- Depends on the velocity of molecules under the influence of a unit driving force
- Provide relation between force and velocity
- Stokes-Einstein Equation: considering the drag on a sphere moving in a continuous fluid

Liquid Mass Diffusivity

Stokes-Einstein Equation

- Describe the diffusion of **colloidal particles** or **large round** molecules through a solvent that behaves as **a continuum** relative to diffusing species

$$D_{AB} = \frac{kT}{6\pi r \mu_B}$$

where:

D_{AB} = Diffusivity of A in dilute solution B (m²/s)

k = Boltzmann constant (1.38 x 10⁻²³ J/K),

T = Absolute temperature (K),

r = Solute particles radius (m),

μ_B = Solvent viscosity (kg/m.s).

Liquid Mass Diffusivity

Wilke-Chang Equation

- For **nonelectrolytes** in an infinitely **dilute** solution
- For solute of small to moderate molecular weight (**< 400**)

$$D_{AB} = \frac{7.4 \times 10^{-8} T (\Phi_B M_B)^{\frac{1}{2}}}{V_A^{0.6} \mu_B}$$

where:

D_{AB} = Diffusivity of A through liquid solvent B (cm²/s)

M_B = Molecular weight of solvent (g/mol),

T = Absolute temperature (K),

V_A = Molecular volume of solute at normal boiling point (cm³/mol)

μ_B = Viscosity of solution (cP)

Φ_B = Association parameter for solvent B

Liquid Mass Diffusivity

Table 24.4 Molecular volumes at normal boiling point for some commonly encountered compounds

Compound	Molecular volume, in cm ³ /g mol	Compound	Molecular volume, in cm ³ /g mol
Hydrogen, H ₂	14.3	Nitric oxide, NO	23.6
Oxygen, O ₂	25.6	Nitrous oxide, N ₂ O	36.4
Nitrogen, N ₂	31.2	Ammonia, NH ₃	25.8
Air	29.9	Water, H ₂ O	18.9
Carbon monoxide, CO	30.7	Hydrogen sulfide, H ₂ S	32.9
Carbon dioxide, CO ₂	34.0	Bromine, Br ₂	53.2
Carbonyl sulfide, COS	51.5	Chlorine, Cl ₂	48.4
Sulfur dioxide, SO ₂	44.8	Iodine, I ₂	71.5

Liquid Mass Diffusivity

Table 24.5 Atomic volumes for complex molecular volumes for simple substances[†]

Element	Atomic volume, in cm ³ /g mol	Element	Atomic volume, in cm ³ /g mol
Bromine	27.0	Oxygen, except as noted below	7.4
Carbon	14.8	Oxygen, in methyl esters	9.1
Chlorine	21.6	Oxygen, in methyl ethers	9.9
Hydrogen	3.7	Oxygen, in higher ethers	
Iodine	37.0	and other esters	11.0
Nitrogen, double bond	15.6	Oxygen, in acids	12.0
Nitrogen, in primary amines	10.5	Sulfur	25.6
Nitrogen, in secondary amines	12.0		

Liquid Mass Diffusivity

Volume Correction

Ring	Correction
Three-membered ring, as ethylene oxide	Deduct 6
Four-membered ring, as cyclobutane	Deduct 8.5
Five-membered ring, as furan	Deduct 11.5
Pyridine	Deduct 15
Benzene ring	Deduct 15
Naphthalene ring	Deduct 30
Anthracene ring	Deduct 47.5

Liquid Mass Diffusivity

Estimation of V_A

$$V_A = 0.285V_c^{1.048}$$

V_c = critical volume of species A (cm³/mol)

Liquid Mass Diffusivity

Association Parameter

Solvent	Φ_B
Water	2.26
Methanol	1.9
Ethanol	1.5
Benzene, ether, heptane, and other unassociated solvents	1.0

Example 1

Estimate the liquid diffusion coefficient of ethanol in a dilute solution of water at 10 °C. Given that the solution viscosity is 1.45 cP.

Liquid diffusivity D_{AB} is not equal to D_{BA}

Liquid Mass Diffusivity

Kayduk and Laudie Equation

- For infinite dilution diffusion coefficients of nonelectrolytes in water

$$D_{AB} = 13.26 \times 10^{-5} \mu_B^{-1.14} V_A^{-0.589}$$

D_{AB} = Mass diffusivity of A through liquid B, water (cm²/s)

μ_B = Viscosity of water (cP)

V_A = Molecular volume of solute at normal boiling point (cm³/mol)

Liquid Mass Diffusivity

Scheibel Equation

- Eliminate association factor in Wilke-Chang relation

$$D_{AB} = \frac{KT}{\mu_B V_A^{\frac{1}{3}}}$$

$$K = 8.2 \times 10^{-8} \left[1 + \left(\frac{3V_B}{V_A} \right)^{\frac{2}{3}} \right]$$

Except

- For benzene as solvent, if $V_A < 2 V_B$, $K = 18.9 \times 10^{-8}$
- For other organic solvent, if $V_A < 2.5 V_B$, $K = 17.5 \times 10^{-8}$

Example 2

Recalculate the diffusion coefficient of ethanol in dilute solution of water in Example 2 if the association parameter is not available.

Liquid Mass Diffusivity

Leffler and Cullinan Equation

- Predicting liquid diffusion coefficient in concentrated solution

$$D_{AB}\mu = (D_{AB}^o\mu_B)^{x_B} (D_{BA}^o\mu_A)^{x_A}$$

D_{AB}^o = Infinite dilution diffusion coefficient

x_A, x_B = Molar fraction of A and B in solution

Liquid Mass Diffusivity

Tyne Equation

- Extrapolate diffusivity at different temperature

$$\frac{D_{AB,T1}}{D_{AB,T2}} = \left(\frac{T_c - T_2}{T_c - T_1} \right)^n$$

T_1 and T_2 are in K

T_c is critical temperature of solvent B

n is constant related to latent heat of vaporization of solvent at its normal boiling point temperature

ΔH_v , (kJ/kmol)	7,900–30,000	30,000–39,000	39,000–46,000	46,000–50,000	>50,000
n	3	4	6	8	10

Example 3

Refer to the data in Appendix Table J2, estimate the diffusivity of acetic acid in water at 400 K and concentration of 0.01 mol/L. Heat of vaporization of water at normal boiling point (100°C, 1 atm) is 40656 J/mol; Critical temperature of water is 647.4K.

Liquid Mass Diffusivity

Nernst Equation

- Determine diffusion coefficient for **dilute** solution of completely ionized **univalent** electrolytes

$$D_{AB} = \frac{2RT}{\left(\frac{1}{\lambda_+^0} + \frac{1}{\lambda_-^0}\right) F_a^2}$$

For univalent ions

D_{AB} = Diffusion coefficient of A in B (cm²/s)

R = Gas constant (8.314 J/K.mol)

T = Absolute temperature (K)

λ_+^0, λ_-^0 = Limiting ionic conductances (A/cm²)

F_a = Faraday constant (96500 Coulombs/mol)

- For polyvalent salt: Replace the constant 2 by $\left(\frac{1}{n^+} + \frac{1}{n^-}\right)$
- n^+ and n^- are the valence of cation and anion

Liquid Mass Diffusivity

Limiting Ionic Conductances In Water (25 °C)

Cation	λ_+^0	Anion	λ_-^0
H ⁺	349.8	OH ⁻	197.6
Li ⁺	38.7	Cl ⁻	76.3
Na ⁺	50.1	Br ⁻	78.3
K ⁺	73.5	I ⁻	76.8
NH ₄ ⁺	73.4	NO ₃ ⁻	71.4

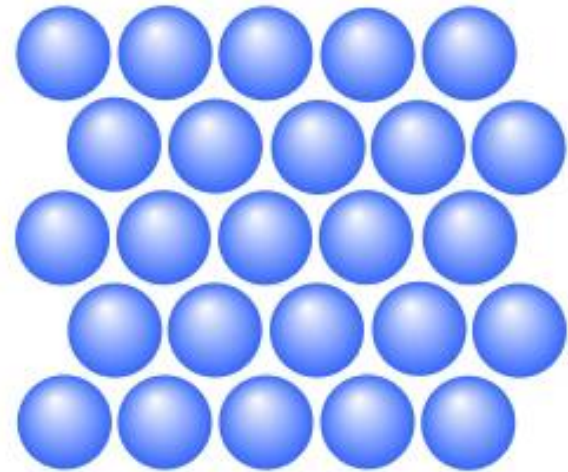
Estimation of Diffusion Coefficient (Solid Diffusivity)

Solid Mass Diffusivity

- Solid diffusion is common in the synthesis of many engineering materials
- Semiconductor: Dopants are introduced into solid silicon to control the conductivity
- Steel: Hardening of steel due to diffusion of carbon and other elements through iron
- Two common diffusion mechanism: vacancy diffusion, interstitial diffusion

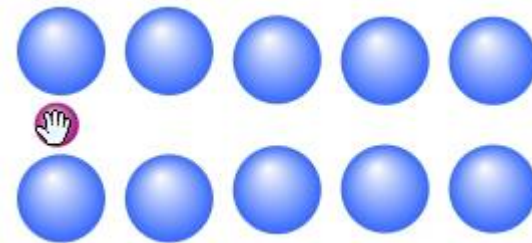
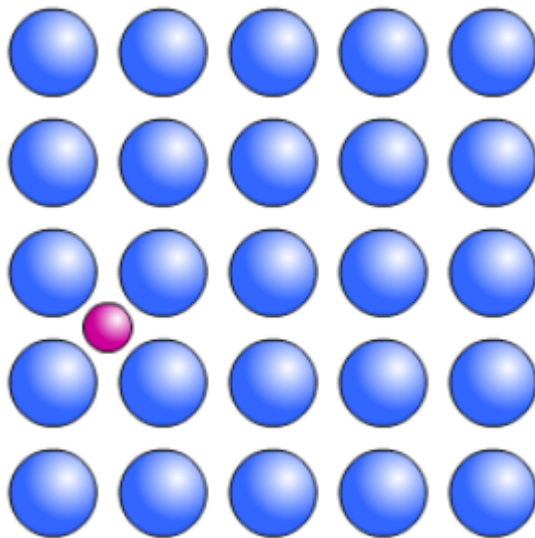
Vacancy Diffusion

- Atom “**jumps**” from a lattice position of solid into neighboring **unoccupied** lattice site/vacancy
- Atom continues to diffuse through solid by a series of jumps
- Atom must “push” other atoms to move from one site to another
- Requires energy
- Causes distortion of lattice
- Diffusion rate depends on number of vacancy & activation energy to exchange



Interstitial Diffusion

- Atom jumps from one **interstitial site** to a neighboring one
- Involve dilation and distortion of lattice
- Faster than vacancy diffusion because the diffusing atoms is usually smaller and there are more interstitial site than vacancy site



Solid Mass Diffusivity

- The diffusivity increases with increasing temperature according to Arrhenius equation

$$D_{AB} = D_o e^{-Q/RT} \quad \Rightarrow \quad \ln(D_{AB}) = -\frac{Q}{R} \frac{1}{T} + \ln(D_o)$$

D_{AB} = Solid diffusion coefficient of A within solid B

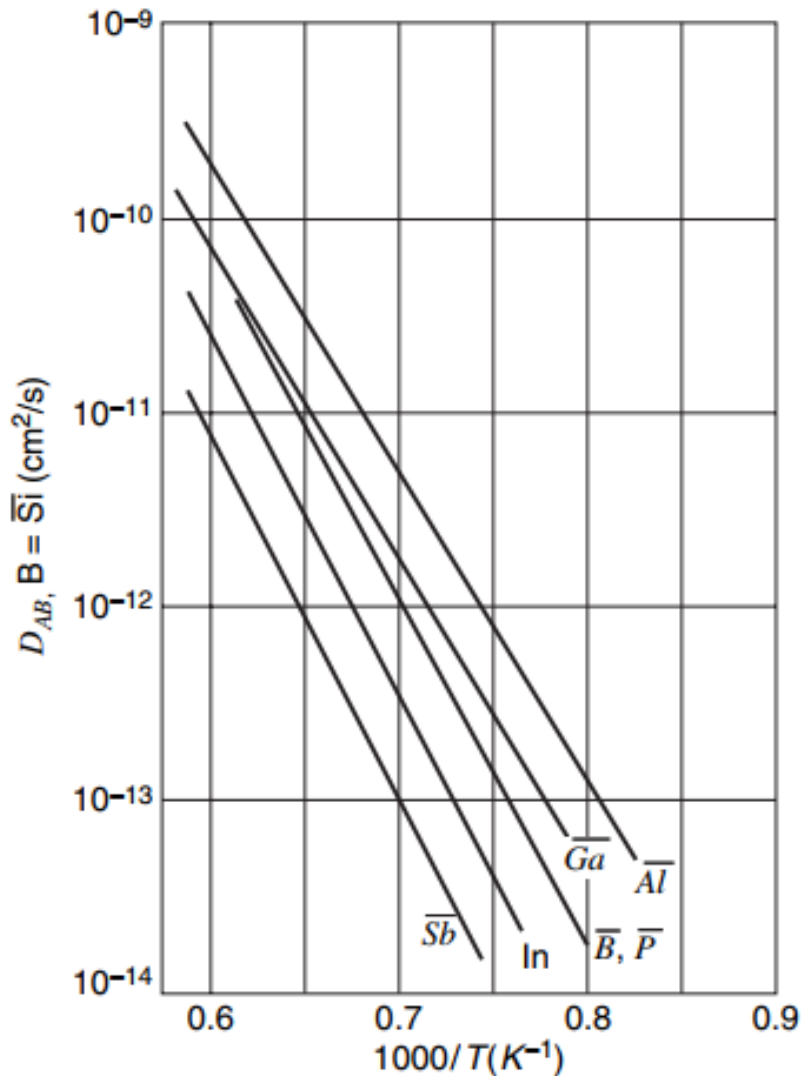
D_o = Proportionality constant

Q = Activation energy (J/mol)

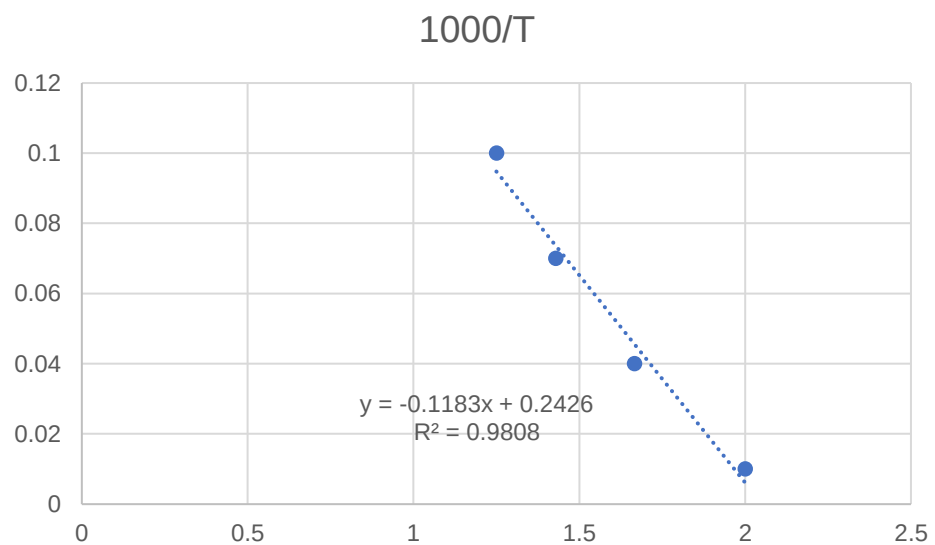
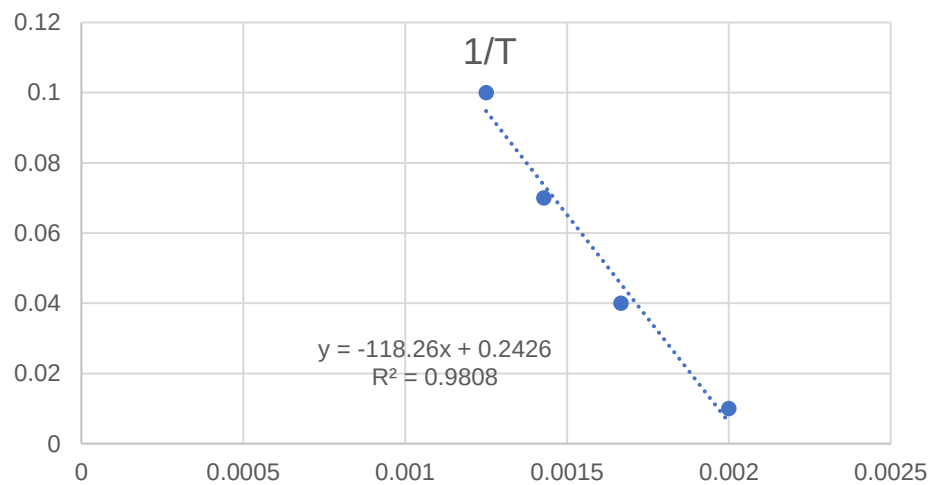
R = Gas constant (8.314 J/mol·K)

T = Absolute temperature (K)

Solid Mass Diffusivity



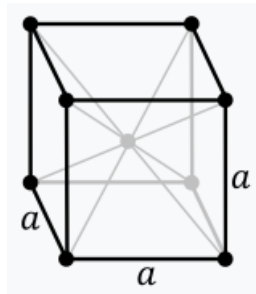
- Diffusion coefficients of substitutional dopants in crystalline silicon
- Can determine Q and D_0



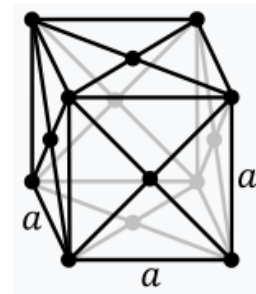
Solid Mass Diffusivity

Table 24.6 Data for self-diffusion in pure metals

Structure	Metal	D_o (mm^2/s)	Q (kJ/mole)
fcc	Au	10.7	176.9
fcc	Cu	31	200.3
fcc	Ni	190	279.7
fcc	Fe(γ)	49	284.1
bcc	Fe(α)	200	239.7
bcc	Fe(δ)	1980	238.5



Body-centered cubic



Face-centered cubic

Solid Mass Diffusivity

Table 24.7 Diffusion parameters for interstitial solutes in iron

Structure	Solute	D_o (mm ² /s)	Q (kJ/mole)
bcc	C	2.0	84.1
bcc	N	0.3	76.1
bcc	H	0.1	13.4
fcc	C	2.5	144.2

Estimation of Diffusion Coefficient (Pore Diffusivity)

Knudsen Diffusion

- Diffusion of gas molecules through a very **small capillary pores**
- Happen when the **pore diameter is smaller** than the mean free path of gas molecules and the **density of gas is low**
- Gas molecules **collide with the wall** more frequently than with each other
- Usually happen in gaseous phase diffusion (mean free path of liquid is small)
- Gas flux is reduced by wall collision

Knudsen Diffusion

Knudsen number

- Measure of the relative important of Knudsen diffusion
- If > 1 , Knudsen diffusion is important

$$Kn = \frac{\lambda}{d_{\text{pore}}} = \frac{\text{mean free path length of the diffusing species}}{\text{pore diameter}}$$

- Knudsen number increase when pressure decrease and temperature increase

Knudsen Diffusion

Knudsen Diffusivity

- Obtained from self-diffusion coefficient
- Replace λ with d_{pore} as molecules is more likely to collide with pore wall

$$D_{AA^*} = \frac{\lambda u}{3} = \frac{\lambda}{3} \sqrt{\frac{8\kappa NT}{\pi M_A}} \quad \longrightarrow \quad D_{KA} = \frac{d_{\text{pore}}}{3} u = \frac{d_{\text{pore}}}{3} \sqrt{\frac{8\kappa NT}{\pi M_A}}$$

Substituting $k = 1.38 \times 10^{-16} \text{ gcm}^2/\text{s}^2\text{K}$ and $N = 6.02 \times 10^{23} \text{ mol}^{-1}$

$$D_{KA} = 4850 d_{\text{pore}} \sqrt{\frac{T}{M_A}}$$

D_{KA} = Knudsen diffusivity (cm^2/s)

d_{pore} = Diameter of pore (cm)

M_A = Molecular weight (g/mol)

T = Temperature (K)

Knudsen Diffusion + Molecular Diffusion

Effective Diffusivity

$$\frac{1}{D_{Ae}} = \frac{1 - \alpha y_A}{D_{AB}} + \frac{1}{D_{KA}}$$

$$\alpha = 1 + \frac{N_B}{N_A}$$

When $\alpha = 0$ ($N_A = -N_B$) or when y_A is close to zero

$$\frac{1}{D_{Ae}} = \frac{1}{D_{AB}} + \frac{1}{D_{KA}}$$

Applied for straight,
cylindrical pores aligned in
parallel array

Knudsen Diffusion + Molecular Diffusion

Effective Diffusivity In Random Pores

- In most porous materials, pores of various diameter are twisted and interconnected
- Diffusion path is **tortuous**

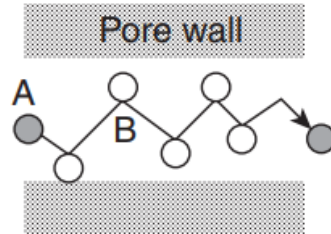
$$D'_{Ae} = \varepsilon^2 D_{Ae}$$

$$\varepsilon = \frac{\text{Volume occupied by pores within the porous solid}}{\text{Total volume of porous solid (solid + pores)}}$$

ε = void fraction

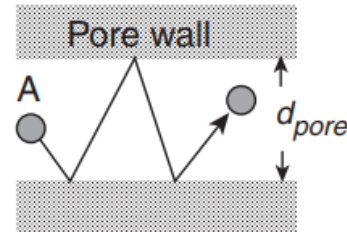
Types of Porous Diffusion

Pure molecular diffusion



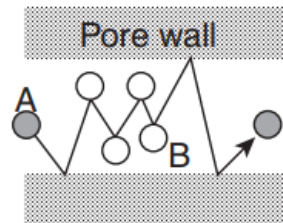
$$D_{AB} = \frac{0.001858 T^{3/2} \left[\frac{1}{M_A} + \frac{1}{M_B} \right]^{1/2}}{P \sigma_{AB}^2 \Omega_D}$$

Pure knudsen diffusion



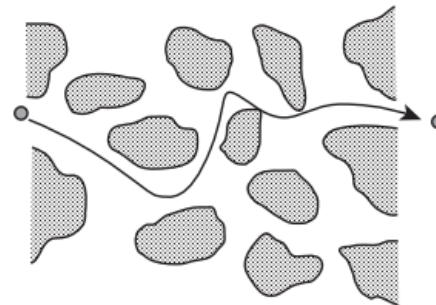
$$D_{KA} = \frac{d_{pore}}{3} \sqrt{\frac{8 \kappa N T}{\pi M_A}}$$

Knudsen + molecular diffusion



$$\frac{1}{D_{Ae}} \cong \frac{1}{D_{AB}} + \frac{1}{D_{KA}}$$

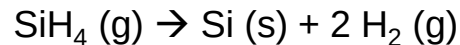
Random porous material



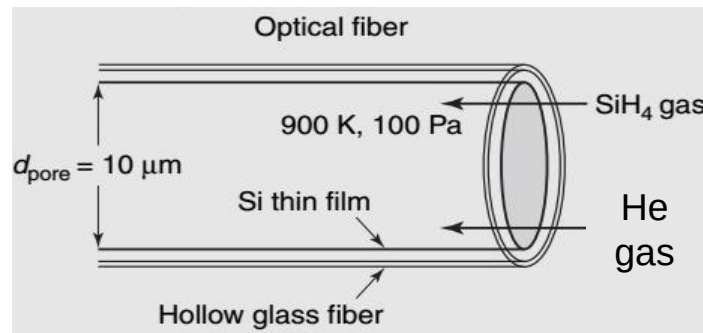
$$D'_{Ae} = \epsilon^2 D_{Ae}$$

Example 4

One step in the manufacture of optical fibers is the chemical vapor deposition of silane (SiH_4) on the inside surface of a hollow glass fiber to form a very thin cladding of solid silicon by reaction:



Typically, the process is carried out at high temperature and very low total system pressure. Optical fibers for high bandwidth data transmission have very small inner pore diameters, typically less than $20 \mu\text{m}$. If the inner diameter of the Si-coated hollow glass fiber is $10 \mu\text{m}$, assess the importance of Knudsen diffusion for SiH_4 inside the fiber lumen at 900 K and 100 Pa total system pressure. Calculate the effective diffusivity of silane in this system. Silane is diluted to $1 \text{ mol}\%$ in the inert carrier gas helium. The binary gas diffusivity of silane in helium at 25°C and 1 atm total system pressure is $0.571 \text{ cm}^2/\text{s}$, with $\sigma_{\text{SiH}_4} = 4.08 \text{ \AA}$ and $\frac{\varepsilon_{\text{SiH}_4}}{k} = 207.6 \text{ K}$. The molecular weight of silane is 32 g/mol .



Hindered Diffusion

- The diffusion of a solute molecule through a **tiny capillary pore filled with liquid solvent**
- As **solute diameter approaches the diameter of pore**, the diffusive transport of solute through the solvent is hindered by the presence of pore and pore wall

Hindered Diffusion

Effective Diffusion Coefficient

$$D_{Ae} = D_{AB}^{\circ} F_1(\varphi) F_2(\varphi)$$

D_{AB}° = Diffusion coefficient of solute A in solvent B at infinite dilution

$F_1(\varphi), F_2(\varphi)$ = correction factor (0 to 1)

Reduced Pore Diameter

$$\varphi = \frac{d_s}{d_{pore}} = \frac{\text{solute molecular diameter}}{\text{pore diameter}}$$

- If $\varphi > 1$, solute too large to enter pore (solute exclusion)
- Use to separate large biomolecules such as protein

Hindered Diffusion

Stearic Partition Coefficient, $F_1(\varphi)$

$$F_1(\varphi) = \frac{\text{flux area available to solute}}{\text{total flux area}} = \frac{\pi (d_{pore} - d_s)^2}{\pi d_{pore}^2} = (1 - \varphi)^2$$

For $0 \leq F_1(\varphi) \leq 1$

Hydrodynamic Hindrance Factor, $F_2(\varphi)$

$$F_2(\varphi) = 1 - 2.104\varphi + 2.09\varphi^3 - 0.95\varphi^5$$

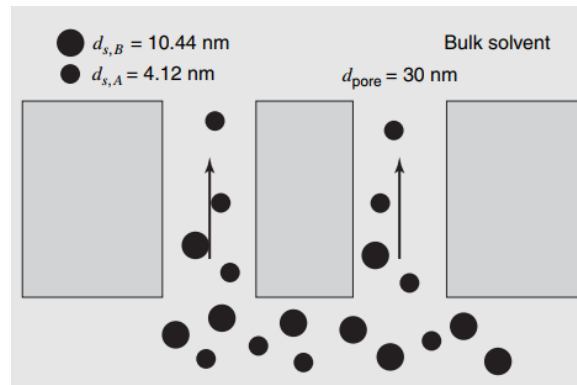
- Assume diffusion of rigid spherical solute in straight cylindrical pore
- Reasonable for $0 \leq (\varphi) \leq 0.6$

Example 5

It is desired to separate a mixture of two industrial enzymes, lysozyme and catalase, in a dilute, aqueous solution (water solvent) by a gel filtration membrane. A mesoporous membrane with cylindrical pores of 30 nm diameter is available. The following separation factor (α) for the process is proposed

$$\alpha = \frac{D_{Ae}}{D_{Be}}$$

Given that the molecular weight and diameter of lysozyme (species A) is 14100 g/mol and 4.12 nm. For catalase (species B), the molecular weight and diameter is 250000 g/mol and 10.44 nm respectively. Determine the separation factor for this process at 298K.



Take Home Message

Stokes-Einstein Equation

$$D_{AB} = \frac{kT}{6\pi r \mu_B}$$

For larger molecules, dilute

Wilke-Chang Equation

$$D_{AB} = \frac{7.4 \times 10^{-8} T (\Phi_B M_B)^{\frac{1}{2}}}{V_A^{0.6} \mu_B}$$

Nonelectrolyte, dilute, small to moderate size ($M_w < 400$)

Kayduk and Laudie Equation

$$D_{AB} = 13.26 \times 10^{-5} \mu_B^{-1.14} V_A^{-0.589}$$

Dilute nonelectrolyte in water

Take Home Message

Scheibel Equation

$$D_{AB} = \frac{KT}{\mu_B V_A^{\frac{1}{3}}}$$

Eliminate association factor

Leffler and Cullinan Equation

$$D_{AB}\mu = (D_{AB}^o\mu_B)^{x_B} (D_{BA}^o\mu_A)^{x_A}$$

For concentrated solution

Tyne Equation

$$\frac{D_{AB,T1}}{D_{AB,T2}} = \left(\frac{T_c - T_2}{T_c - T_1} \right)^n$$

Predict diffusivity at different temperature

Take Home Message

Solid Diffusion

$$D_{AB} = D_o e^{-Q/RT}$$

Vacancy & Interstitial diffusion

Knudsen Diffusivity

$$D_{KA} = 4850 d_{pore} \sqrt{\frac{T}{M_A}}$$

Mean free path larger than pore size

Take Home Message

Effective Diffusivity Equation

$$\frac{1}{D_{Ae}} = \frac{1 - \alpha y_A}{D_{AB}} + \frac{1}{D_{KA}}$$

Straight, cylindrical pore

$$\frac{1}{D_{Ae}} = \frac{1}{D_{AB}} + \frac{1}{D_{KA}}$$

Effective Diffusivity In Random Pores

Pore with tortuous path

$$D'_{Ae} = \varepsilon^2 D_{Ae}$$

Take Home Message

Hindered Diffusion

$$D_{Ae} = D_{AB}^{\circ} F_1(\varphi) F_2(\varphi)$$

Pore filled with solvent, more significant if the size of molecule approaching pore diameter